

```

root@ip-172-31-21-31:/# lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                7:0      0 27.6M  1 loop /snap/amazon-ssm-agent/11797
loop1                7:1      0   74M  1 loop /snap/core22/2163
loop2                7:2      0 50.9M  1 loop /snap/snapd/25577
nvme0n1              259:0     0    8G  0 disk
├─nvme0n1p1          259:1     0    7G  0 part /
├─nvme0n1p14         259:2     0    4M  0 part
├─nvme0n1p15         259:3     0 106M  0 part /boot/efi
└─nvme0n1p16         259:4     0 913M  0 part /boot
nvme3n1              259:7     0   14G  0 disk
root@ip-172-31-21-31:/#

```

```

root@ip-172-31-21-31:/# pvs
root@ip-172-31-21-31:/# vgs
root@ip-172-31-21-31:/# lvs
root@ip-172-31-21-31:/# |

```

```

root@ip-172-31-21-31:/# df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        6.8G  1.8G  5.0G  27% /
tmpfs            458M    0  458M   0% /dev/shm
tmpfs            183M  900K  182M   1% /run
tmpfs            5.0M    0   5.0M   0% /run/lock
efivarfs         128K  3.6K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p16  881M   89M  730M  11% /boot
/dev/nvme0n1p15  105M   6.2M   99M   6% /boot/efi
tmpfs            92M   12K   92M   1% /run/user/1000
root@ip-172-31-21-31:/# |

```

An EVS Volume of 14GB was attached to the instance and a physical volume of 14 GB is created .

```

lvm> pvcreate /dev/nvme3n1
Physical volume "/dev/nvme3n1" successfully created.
lvm> pvdisplay
"/dev/nvme3n1" is a new physical volume of "14.00 GiB"
--- NEW Physical volume ---
PV Name                /dev/nvme3n1
VG Name
PV Size                14.00 GiB
Allocatable            NO
PE Size                0
Total PE               0
Free PE                0
Allocated PE           0
PV UUID                bYoYzD-Z580-cUUC-o5Qc-EvpA-N7rs-8DMwTb

lvm>

```

A volume group is created

```
lvm> vgcreate batch10_vg /dev/nvme3n1
Volume group "batch10_vg" successfully created
lvm> vdisplay
--- Volume group ---
VG Name                batch10_vg
System ID
Format                 lvm2
Metadata Areas         1
Metadata Sequence No   1
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                 0
Max PV                 0
Cur PV                 1
Act PV                 1
VG Size                 <14.00 GiB
PE Size                 4.00 MiB
Total PE                3583
Alloc PE / Size         0 / 0
Free PE / Size          3583 / <14.00 GiB
VG UUID                 9EtPe2-Qjxb-kL51-P2R4-Grg4-1eyA-dmyF1A

lvm>
```

```
lvm> lvcreate -L 2g batch10_vg -n app-data
Logical volume "app-data" created.
lvm> lvdisplay
--- Logical volume ---
LV Path                /dev/batch10_vg/app-data
LV Name                 app-data
VG Name                 batch10_vg
LV UUID                 KYID1D-lheb-aLR3-u1pt-Ij0j-gceX-7QCv2o
LV Write Access         read/write
LV Creation host, time  ip-172-31-21-31, 2026-02-20 13:18:30 +0000
LV Status                available
# open                  0
LV Size                 2.00 GiB
Current LE              512
Segments                1
Allocation              inherit
Read ahead sectors      auto
- currently set to     256
Block device            252:2

lvm>
```

Created one more LV name app-logs

```
lvcreate -L 5g batch10_vg -n app-logs
Logical volume "app-logs" created.
lv> lvs
--- Logical volume ---
LV Path                /dev/batch10_vg/app-data
LV Name                 app-data
VG Name                 batch10_vg
LV UUID                 KYID1D-1heb-aLR3-u1pt-Ij0j-gceX-7QCv2o
LV Write Access         read/write
LV Creation host, time ip-172-31-21-31, 2026-02-20 13:18:30 +0000
LV Status                available
# open                  0
LV Size                 2.00 GiB
Current LE              512
Segments                1
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            252:2

--- Logical volume ---
LV Path                /dev/batch10_vg/app-logs
LV Name                 app-logs
VG Name                 batch10_vg
LV UUID                 OPTaXm-h42t-ex0o-qy1n-c2SK-mW45-ZMgtxk
LV Write Access         read/write
LV Creation host, time ip-172-31-21-31, 2026-02-20 13:20:50 +0000
LV Status                available
# open                  0
LV Size                 5.00 GiB
Current LE              1280
Segments                1
Allocation               inherit
Read ahead sectors      auto
- currently set to     256
Block device            252:3

lv> |
```

```

root@ip-172-31-21-31:/# mkdir /mnt/app-data
root@ip-172-31-21-31:/# mkdir /mnt/app-logs
root@ip-172-31-21-31:/# mkfs -t ext4 /dev/batch10_vg/app-data
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 524288 4k blocks and 131072 inodes
Filesystem UUID: 43da8828-a37b-4800-be2e-c00396214023
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-21-31:/# mkfs -t ext4 /dev/batch10_vg/app-logs
mke2fs 1.47.0 (5-Feb-2023)
Creating filesystem with 1310720 4k blocks and 327680 inodes
Filesystem UUID: 8712b715-2e8b-49e4-be6c-624ef0ea0a92
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

root@ip-172-31-21-31:/#

```

```

root@ip-172-31-21-31:/# mount /dev/batch10_vg/app-data /mnt/app-data/
root@ip-172-31-21-31:/# mount /dev/batch10_vg/app-logs /mnt/app-logs/
root@ip-172-31-21-31:/#

```

```

root@ip-172-31-21-31:/# df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/root	6.8G	1.8G	5.0G	27%	/
tmpfs	458M	0	458M	0%	/dev/shm
tmpfs	183M	908K	182M	1%	/run
tmpfs	5.0M	0	5.0M	0%	/run/lock
efivarfs	128K	3.6K	120K	3%	/sys/firmware/efi/efivars
/dev/nvme0n1p16	881M	89M	730M	11%	/boot
/dev/nvme0n1p15	105M	6.2M	99M	6%	/boot/efi
tmpfs	92M	12K	92M	1%	/run/user/1000
/dev/mapper/batch10_vg-app--data	2.0G	24K	1.8G	1%	/mnt/app-data
/dev/mapper/batch10_vg-app--logs	4.9G	24K	4.6G	1%	/mnt/app-logs

```

root@ip-172-31-21-31:/# |

```

```

lvm> lvextend -L +2g /dev/batch10_vg/app-data
  Size of logical volume batch10_vg/app-data changed from 2.00 GiB (512 extents) to 4.00 GiB (1024 extents).
  Logical volume batch10_vg/app-data successfully resized.
lvm>

```

Resize2fs doesn't work on lvm, needs to run on the terminal

```
root@ip-172-31-21-31:/# resize2fs /dev/batch10_vg/app-data
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/batch10_vg/app-data is mounted on /mnt/app-data; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 1
The filesystem on /dev/batch10_vg/app-data is now 1048576 (4k) blocks long.

root@ip-172-31-21-31:/# df -h /mnt/app-data/
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/batch10_vg-app--data  3.9G    24K   3.7G   1% /mnt/app-data
root@ip-172-31-21-31:/#
```